Constant-Wind Zermelo Navigation Beyond the Randers Chamber

Route-A source-local certificate HCS-C305

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Abstract

For every finite $d \geq 1$, constant wind $W \in \mathbb{R}^d$, control cap $c \geq 0$, and target, we give the complete minimum-time atlas for $\dot{x} = W + u$, $|u| \leq c$. A single theorem covers weak, critical, and strong wind, including exact reachability, value formulas, and all attainable times.

1 The all-chamber value theorem

Consider measurable controls u satisfying $|u(t)| \leq c$ a.e. and

$$\dot{x} = W + u, \qquad x(0) = 0. \tag{1}$$

For a target y, write

$$w = |W|, \qquad p = W \cdot y, \qquad r = |y|, \qquad a = w^2 - c^2, \qquad D = p^2 - ar^2.$$

Theorem 1 (Complete constant-wind atlas). *At exact time $t \geq 0$, the reachable set is*

$$\mathcal{R}(t) = Wt + ct\overline{B}. \tag{2}$$

For $y \neq 0$, minimum time and all attainable times are as follows.

1. *If $w < c$, every y is reachable,*

$$T(y) = \frac{\sqrt{p^2 + (c^2 - w^2)r^2} - p}{c^2 - w^2}, \qquad \mathcal{I}(y) = [T, \infty). \tag{3}$$

2. *If $w = c > 0$, reachability is equivalent to $p > 0$, and then*

$$T(y) = \frac{r^2}{2p}, \qquad \mathcal{I}(y) = [T, \infty). \tag{4}$$

3. *If $w > c$, reachability is equivalent to $p > 0$ and $D \geq 0$. Then*

$$T_{\mp}(y) = \frac{p \mp \sqrt{D}}{w^2 - c^2}, \qquad T(y) = T_-(y), \qquad \mathcal{I}(y) = [T_-, T_+]. \tag{5}$$

For every nonzero reachable target the time-optimal control is unique up to null sets and constant:

$$u_*(s) = y/T(y) - W, \qquad |u_*| = c. \tag{6}$$

Proof. Integrating (1) shows that control averages fill $c\overline{B}$, proving (2). Thus y is attainable at t exactly when

$$|y - Wt|^2 \leq c^2 t^2 \quad \Longleftrightarrow \quad f(t) = at^2 - 2pt + r^2 \leq 0. \tag{7}$$

For $y \neq 0$, $f(0) > 0$. If $a < 0$, its roots have opposite signs and the positive root is (3); the inequality holds thereafter. If $a = 0$, it is linear and crosses zero at positive time exactly when $p > 0$, giving (4). If $a > 0$, it has a nonnegative-time sublevel interval exactly when $p > 0$ and $D \geq 0$. Both roots are then positive and the sublevel set is precisely (5); in particular the minimum is the smaller root, not T_+.

At first contact, $|y - WT| = cT$. Any optimal control obeys

$$\left| T^{-1} \int_0^T u(s)\, ds \right| = c \leq T^{-1} \int_0^T |u(s)|\, ds \leq c.$$

Equality in the strictly convex Euclidean triangle inequality forces the control to equal $y/T - W$ a.e. For $c = 0$ this conclusion is immediate. $\square$

Source lineage

Zermelo's 1931 paper is the historical owner token for the navigation problem. Bao, Robles, and Shen provide navigation/Randers geometric context. The citations establish lineage, not novelty or priority.

References

[1] E. Zermelo, "Über das Navigationsproblem bei ruhender oder veränderlicher Windverteilung," *ZAMM* 11 (1931), 114–124. DOI: 10.1002/zamm.19310110205.

[2] D. Bao, C. Robles, and Z. Shen, "Zermelo navigation on Riemannian manifolds," *J. Differential Geom.* 66 (2004), 377–435. DOI: 10.4310/jdg/1098137838.